

Dr Jivraj N Mehta Government Polytechnic-Amreli
EC/ICT

SEMESTER:-3

REMEDIAL EXAM

WINTER-2025

Course Code & Name : DI03032031 (Data Structure and Applications)

Duration : 11:30 AM to 12:30 PM

Date:- 21-11-2025

Maximum Marks : 30

Instructions:

1. Attempt ALL questions.
2. Make suitable assumptions wherever necessary.
3. Use of Programmable Calculator or any similar Mathematical aid is strictly prohibited.
4. Numbers on right side of question indicates maximum marks of that question

CO	Statement	Mark
CO1	Comprehend the basic concepts of data structures, analysis terms, OOPs concepts, and recursive functions.	10
CO2	Apply basic operations on stack, queue and linked list data structures	15
CO5	Illustrate algorithms to insert, delete and searching a node in tree. R,U.	5

- Q1 Define List. Write various operations that can be performed on List. 3 CO1 R
- Q2 Write an algorithm for PUSH and POP operation of stack in python 5 CO2 R
- Q3 Differentiate between Stack and Queue. 5 CO2 U
- OR
- Q3 Explain concept of circular queue. 5 CO2 U
- Q4 Define Time and space complexity 3 CO1 A
- Q5 Convert following equation from infix to postfix using Stack. 5 CO2 A
- ✓ $A * (B + C) - D / (E + F) +$
- Q6 Define following terms regarding Tree data structure: 5 CO5 U
1. Forest
 2. Root node
 3. Leaf node
- OR
- Define following terms regarding Tree data structure: 5
1. In-degree
 2. Out-degree
 3. Depth
- Q6 5 CO5 U
- Q7 Explain class, attributes, object and class method in python with suitable example. 4 CO1 U
- OR
- Q7 Define linear and non linear data structure and give its examples 4 CO1 U

$D / E F +$

$D / E F +$

$A * (B + C) - D E F + /$

$A * (B C +) - D E F + /$

$A * (B C +) - D E +$

$(A B C + *) - D E F + /$

$A B C + * D E F + / -$



SEMESTER:-3

Course Code & Name : DI03032021 (Java Programming)

Duration : 11:30 AM to 12:30 PM

Date:- 20-11-2025

Maximum Marks : 30

Instructions:

1. Attempt ALL questions.
2. Make suitable assumptions wherever necessary.
3. Use of Programmable Calculator or any similar Mathematical aid is strictly prohibited.
4. Numbers on right side of question indicates maximum marks of that question

CO	Statement	Mark
CO1	Write simple Java programs using basic syntax, data types, and control structures.	6
CO2	Apply object-oriented programming concepts to solve real-world problems.	14
CO3	Develop Java applications using inheritance, interfaces, and packages.	10

- Q1 List out the rules to name an identifier in Java with valid and invalid examples 3 CO1 R
- Q2 List out different types of operators in Java. Explain Arithmetic and Logical Operators in detail. 3 CO1 U
- OR
- Q2 List out various Primitive data types in Java. 3 CO1 U
- Q3 Write syntax of Java for loop statement. Develop a Java program to print number between 1 to 50. 4 CO2 U
- Q4 List the differences between Procedure-Oriented Programming (POP) and Object-Oriented Programming (OOP). 5 CO2 U
- Q5 Define Constructor. List types of Constructors. Develop a java code to explain Parameterized constructor 5 CO2 A
- OR
- Q5 List out different types of inheritance and explain any one with JAVA code. 5 CO2 A
- Q6 Explain super and static keyword with example 5 CO3 A
- Q7 Define package and write the syntax to create a package with example. 5 CO3 U
- OR
- Q7 List out any four differences between method overloading and overriding. 5 CO3 U

SEMESTER-3

Course Code & Name : DI03032021 (Java Programming)

Duration : 11:30 AM to 12:30 PM

Date:- 25-09-2025

Maximum Marks : 30

Instructions:

1. Attempt ALL questions.
2. Make suitable assumptions wherever necessary.
3. Use of Programmable Calculator or any similar Mathematical aid is strictly prohibited.
4. Numbers on right side of question indicates maximum marks of that question

CO	Statement	Mark
CO1	Write simple Java programs using basic syntax, data types, and control structures.	3
CO2	Apply object-oriented programming concepts to solve real-world problems.	13
CO3	Develop Java applications using inheritance, interfaces, and packages.	14

Q1 Write a program in Java to reverse the digits of a number for number having three digits. Like reverse of 478 is 874. 3 CO1 R

2 Q2 List out basic concepts of OPP. Explain any one in details. 3 CO2 U

Q2 Write a program in JAVA to add two 3*3 matrices. 3 CO2 U

Q3 Explain super keyword with example. 4 CO3 U

Q4 Explain different access controls in Java. 5 CO3 U

Q5 Explain any two String buffer and two String class methods with suitable example. 5 CO2 A

Q5 Define: Method Overriding. List out Rules for method overriding. Write a java program that implements method overriding. 5 CO2 A

Q6 Define abstract class in JAVA with example. 5 CO2 A

Q7 Define package and write the syntax to create a package with example. 5 CO3 U

Q7 Write types of Inheritance & Write a program in Java to demonstrate multiple inheritance. 5 CO3 U

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SEMESTER EXAMINATION-2025

MID EXAM

Semester : 3rd
Date: 23/09/2025

Sub.: Principle of Electronic Communication
Time : 11.30 to 12.30

Sub. Code: DI03000121
Max. Marks: 30

Instructions:

- English version is considered to be Authentic.
- Figure to the right indicate full marks.

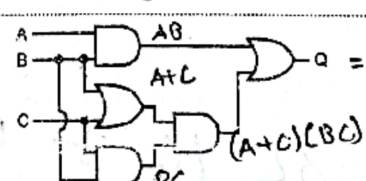
QUE. NO.	QUESTIONS	MARKS	CO	BL
Q.1	Attempt any TWO questions. (કોઈપણ બે પ્રશ્નો ના જવાબ આપો)	Total Marks - 10		
1A	Draw and explain block diagram of Digital communication system	05	CO1	L1
1B	Write electromagnetic wave spectrum, frequency bands and their applications	05	CO1	L2
1C	Draw following waveforms in time domain and frequency domain (i) Square wave (ii) Triangular wave (iii) Sawtooth wave (iv) Sinusoidal wave	05	CO1	L2
1D	What is noise? Classify Noise and describe each one.	05	CO1	L2
Q.2	Attempt any TWO questions. (કોઈપણ બે પ્રશ્નો ના જવાબ આપો)	Total Marks - 10		
2A	Define (i) Sensitivity (ii) Selectivity (iii) Fidelity (iv) Image Frequency rejection	05	CO2	L1
2B	Draw and Explain block diagram of Superhetrodyne AM receiver and their different blocks.	05	CO2	L2
2C	Derive mathematical equation of Amplitude Modulated wave.	05	CO2	L2
2D	What is modulation? Why it is needed? Draw and Explain Amplitude Modulation.	05	CO2	L2
Q.3	Attempt any TWO questions. (કોઈપણ બે પ્રશ્નો ના જવાબ આપો)			
3A	Draw and explain frequency modulation. Derive mathematical equation of FM signal.	05	CO3	L1
3B	Draw and explain PLL (phase lock loop) demodulator.	05	CO3	L2
3C	Draw and explain Foster-Seeley Discriminator.	05	CO3	L2
3D	Compare AM, FM and PM	05	CO3	L2

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MID SEMESTER EXAM WINTER 2025

Semester : 3rd
 Date: 22/09/2025

Sub.: Digital Logic Design
 Time : 11:30 PM to 12:30 PM

Sub. Code: DI03000111
 Max. Marks: 30

QUE. NO.	QUESTIONS	MARKS	CO
Q.1A	Attempt any two questions. <i>(53)10</i>		
(1)	Convert: $(110101)_2 = (\text{---})_{10} = (\text{---})_8 = (\text{---})_{16}$	03	CO1
(2)	Divide $(1111101)_2$ with $(101)_2$	03	CO1
(3)	Perform subtraction using 2's complement method (i) $(110110)_2 - (101010)_2$	03	CO1
Q.1B	Attempt any one questions.		
(1)	Perform: (i) $(11101101)_2 + (10101000)_2$ (ii) $(11011)_2 \times (1010)_2$	04	CO1
(2)	Solve $(8E)_{16} + (100)_2 = (\text{---})_{10}$	04	CO1
Q.2A	Attempt any two question.		
(1)	Draw EX-OR GATE using NAND Gate only.	03	CO2
(2)	State De-Morgan's theorems and prove it.	03	CO2
(3)	 $Q = AB + (A+C)(BC)$ find equation for Q.	03	CO2
Q.2B	Attempt any one question.		
(1)	Simplify the Boolean expression using algebraic method $Y = A + B[AC + (B + \bar{C})D]$	04	CO2
(2)	Simplify with the help of K-Map for expression $f(A,B,C,D) = \sum m(1,2,6,7,8,13,14,15) + d(3,5,12)$	04	CO2
Q.3A	Attempt any two question.	Total Marks - 06	
(1)	Explain half adder using logic circuit and truth table	03	CO3
(2)	Explain 1:4 Demultiplexer with logic circuit and truth table	03	CO3
(3)	Draw the circuit of 4 bit Binary to gray code converter.	03	CO3
Q.3B	Attempt any one question.	Total Marks - 04	
(1)	Describe 8:1 Multiplexer with truth table, equation and circuit diagram.	04	CO3
(2)	Explain full Subtractor using logic circuit and truth table.	04	CO3

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128093168921
 11101101
 $256 + 128 + 64 + 32 + 8 + 4 + 2 = 237$
 $1 + 1$
 168
 405

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ICT ENGINEERING DEPARTMENT
Mid Sem Examination WINTER 2025

Semester : 3rd

Sub.: COMPUTER NETWORKING

Sub. Code: DI03000131

Date: 24/09/2025

Time : 11:30 TO 12:30 PM

Max. Marks: 30

Instructions:

- English version is considered to be Authentic.
- Figure to the right indicate full marks.

QUE. NO.	QUESTIONS	MARKS	CO
Q.1	Attempt any one questions.	Total Marks - 14	
✓ (1)	Draw the diagram of TCP/IP suite and Explain the function of Each Layer.	07	CO1
	OR		
(1)	Draw and Explain the OSI Model for Computer Network	07	CO1
(2)	Compare hub, switch and Router.	07	CO2
	OR		
✓ (2)	Write short note on hub and Router	07	CO2
Q.2	Attempt any two questions.	Total Marks - 10	
✓ (1)	Write Short Note on LAN and MAN	05	CO1
(2)	Explain Types of Different Transmission Media.	05	CO2
(3)	Draw and Explain the Function of Network Address Translation (NAT).	05	CO3
✓ (4)	Differentiate IPV4 and IPV6	05	CO3
Q.3	Attempt any one Following question.	Total Marks - 06	
(1)	Explain DHCP And ARP Protocol in Detail.	06	CO3
	OR		
✓ (1)	Explain Classfull IP Addressing system.	06	CO3

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SEMESTER:-3

1st MIDSEM EXAM

WINTER-2025

Course Code & Name : DI03032031 (Data Structure and Applications)

Duration : 11:30 AM to 12:30 PM

Date:- 26-09-2025

Maximum Marks : 30

Instructions:

1. Attempt ALL questions.
2. Make suitable assumptions wherever necessary.
3. Use of Programmable Calculator or any similar Mathematical aid is strictly prohibited.
4. Numbers on right side of question indicates maximum marks of that question

CO	Statement	Mark
C01	Comprehend the basic concepts of data structures, analysis terms, OOPs concepts, and recursive functions.	10
C02	Apply basic operations on stack, queue and linked list data structures	10
C05	Illustrate algorithms to insert, delete and searching a node in tree. R,U,	10

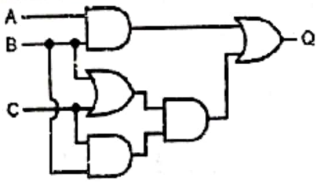
- Q1 Explain Linear and Non Linear Data structure in Python with examples. 3 C01 R
- Q2 Define following terms regarding Tree data structure:
1. In-degree
2. Out-degree
3. Depth 5 C05 R
- Q3 Explain different operations on stack with any one algorithm. 5 C02 U
OR
- Q3 Explain different operations on queue with any one algorithm. 5 C02 U
- Q4 Define List. Write various operations that can be performed on List. 3 C01 A
- Q5 Convert following algebraic expression into postfix notation using Stack 5 C02 A
i) $(a*b)*(c^d(d+e)-f)$
ii) $a-b/(c*d/e)$
- Q6 Draw Binary search tree for 8, 4, 12, 2, 6, 10, 14, 1, 3, 5 and write In-order traversal for the tree. 5 C05 U
OR
- Q6 Draw Binary search tree for 8, 4, 12, 2, 6, 10, 14, 1, 3, 5 and write post-order traversal for the tree. 5 C05 U
- Q6 Write a Python class to represent a Cricketer. 4
The class contains the name of the cricketer, team name and run as the data members.
The member functions are as follows: to initialize the data members, to set run and display run. C01 U
- Q7 Explain concept of recursive function with any one suitable code in python. 4 C01 U
OR

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ICT ENGINEERING DEPARTMENT
REMID SEMESTER EXAM WINTER 2025

Semester : 3rd
 Date: 17/10/2025

Sub.: Digital Logic Design
 Time: 11:30 PM to 12:30 PM

Sub. Code: DI03000111
 Max. Marks: 30

QUE. NO.	QUESTIONS	MARKS	CO
Q.1A	Attempt any two questions.		
(1)	Convert: $(110101)_2 = (\quad)_{10} = (\quad)_8 = (\quad)_{16}$	03	CO1
(2)	Divide $(1111101)_2$ with $(101)_2$	03	CO1
(3)	Perform subtraction using 1's complement method (i) $(110110)_2 - (101010)_2$	03	CO1
Q.1B	Attempt any one questions.		
(1)	Perform: (i) $(11101101)_2 + (10101000)_2$ (ii) $(11011)_2 \times (1010)_2$	04	CO1
(2)	Solve $(E)_{16} + (F)_2 = (\quad)_{10}$	04	CO1
Q.2A	Attempt any two question.		
(1)	Draw EX-NOR GATE using NAND Gate only.	03	CG2
(2)	State De-Morgan's theorems and prove it.	03	CO2
(3)	 find equation for Q.	03	CO2
Q.2B	Attempt any one question.		
(1)	Prove that: $(A + C)(A.B + C) = 0$	04	CO2
(2)	Simplify with the help of K-Map for expression $f(X,Y,Z,W) = \sum m(1,2,3,5,7,12,14,15) + d(6,8,13)$	04	CO2
Q.2A	Attempt any two question.	Total Marks - 06	
(1)	Explain half subtractor using logic circuit and truth table	03	CO3
(2)	Explain 1:4 Demultiplexer with logic circuit and truth table	03	CO3
(3)	Draw the circuit of 4 bit Gray to Binary code converter.	03	CO3
Q.2B	Attempt any one question.	Total Marks - 04	
(1)	Describe 3:1 Multiplexer with truth table, equation and circuit diagram.	04	CO3
(2)	Explain full adder using logic circuit and truth table.	04	CO3